Financial Mathematics (MATH35400/MATHM5400)

Additional Example – 25 February 2026 [1]

1. (2022 exam, Q2 (a)) The current price for Bibi shares is £50 per share. A put option on Bibi shares maturing in exactly 9 months with a strike price of £40 is currently trading at £14. The price of the corresponding call option with the same strike price and same maturity is currently £26.

   (a) The put-call parity states that if there are no arbitrage opportunities, the following formula holds:
   $$S_t + P_t - C_t = Ke^{-r(T-t)}.$$
   State the definition of each of the quantities appearing in this formula and use it to show that when $r = 0.05$ and time t is measured in years, there exist arbitrage opportunities in the model above.

   (b) Describe explicitly a trading strategy which will give riskless profit.

Solution:

(a) r is the (time-continuous) interest rate, assumed constant constant during the entire interval $[0, T]$, S_t is the price of an asset at time point t and C_t and P_t are the time t prices of European call and put options on the same asset with strike price K and maturity T.

In this particular situation the left hand side equals $50 + 14 - 26 = 38$, but the right-hand side equals 38.53. Hence there exist arbitrage opportunities.

(b) Today the arbitrageur would borrow £38 from the bank and short sell 1 call option and use the £64 to buy 1 unit of the underlying asset and 1 put option.

9 months later the arbitrageur owes the bank £39.45.

If the stock price $S_T < 40$, then we use our put option to sell our stock for £40 and repay our loan, making a profit of £0.55 since the call option is worthless – if the buyer of the call option decides to exercise we just buy another stock and sell it to them making even more profit.

If the stock price is $S_T \geq 40$, then the short seller of the call option will want to exercise the option and therefore we need to return the stock to them and the pay us £40 which we use to repay our loan, making again a profit of £0.55. The put option just expires.

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2. (2022 exam, Year 4, Q2 (b)) Assuming the availability of a bank account which pays interest at rate r, prove by no-arbitrage that

$$C_t^a + K \geq S_t + P_t^a,$$

for any $t \leq T$, where C_t^a and P_t^a are the time t prices of respectively American call and put options on the same underlying asset with exercise price K and expiry date T, and S_t is the price of the underlying asset.

Solution: Assume to the contrary that $C_t^a + K < S_t + P_t^a$. Then selling the security and put and buying the call produces the cash flow $S_t + P_t^a - C_t^a > K$. Invest this amount in the bank. If the owner of the American put chooses to exercise it at time $s \in [t, T]$, then we need to buy the security from them for K and then return the security to the lender of the stock. The net balance of the investment is

$$(S_t + P_t^a - C_t^a)e^{r(s-t)} - K > Ke^{r(s-t)} - K \geq 0.$$

If the American put expires out of the money, we exercise the call option to close the short position in the security at time T. The net balance of the investment is

$$(S_t + P_t^a - C_t^a)e^{r(T-t)} - K > Ke^{r(T-t)} - K > 0.$$

Thus the investor receives a non-negative profit in either case, violating the principle of no arbitrage. [Similar answer acceptable.]
